

Homework #7

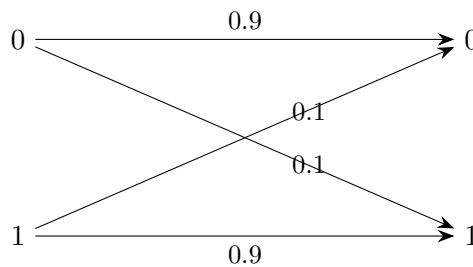
The Channel Coding Theorem

Introduction to Information Theory 2026 · EC5203 / EC4310 / MM4450 · Prof. Heung-No Lee · GIST

How to submit. Put your solution in the `submission/` folder of your assignment repository: a single PDF (`submission/hw7.pdf` — scanned handwriting is fine if legible), or Markdown with L^AT_EX (`submission/hw7.md`). Commit and push before the deadline (23:59 KST); the last commit before the deadline is graded. If you worked as a group (max 3), list all members at the top — each member still submits in their own repository.

Questions. Light ones in the AI TA chat on the course homepage; rigorous derivations and proofs on the course GitHub Q&A (mention @claude). Before the deadline you get hints, never solutions.

Problem HW7.1 (C&T 7.15) *Jointly typical sequences.* As we did in Problem 3.13 for the typical set for a single random variable, we will calculate the jointly typical set for a pair of random variables connected by a binary symmetric channel, and the probability of error for jointly typical decoding for such a channel.



We consider a binary symmetric channel with crossover probability 0.1. The input distribution that achieves capacity is the uniform distribution [i.e., $p(x) = (\frac{1}{2}, \frac{1}{2})$], which yields the joint distribution $p(x, y)$ for this channel given by

$X \backslash Y$	0	1
0	0.45	0.05
1	0.05	0.45

The marginal distribution of Y is also $(\frac{1}{2}, \frac{1}{2})$.

- (a) Calculate $H(X)$, $H(Y)$, $H(X, Y)$, and $I(X; Y)$ for the joint distribution above.
- (b) Let $X_1, X_2, \dots, X_n$ be drawn i.i.d. according the Bernoulli($\frac{1}{2}$) distribution. Of the 2^n possible input sequences of length n , which of them are typical [i.e., member of $A_\epsilon^{(n)}(X)$ for $\epsilon = 0.2$]? Which are the typical sequences in $A_\epsilon^{(n)}(Y)$?
- (c) The jointly typical set $A_\epsilon^{(n)}(X, Y)$ is defined as the set of sequences that satisfy equations (7.35–7.37). The first two equations correspond to the conditions that x^n and y^n are in $A_\epsilon^{(n)}(X)$ and $A_\epsilon^{(n)}(Y)$, respectively. Consider the last condition, which can be rewritten to state that $-\frac{1}{n} \log p(x^n, y^n) \in (H(X, Y) - \epsilon, H(X, Y) + \epsilon)$. Let k be the number of places

in which the sequence x^n differs from y^n (k is a function of the two sequences). Then we can write

$$p(x^n, y^n) = \prod_{i=1}^n p(x_i, y_i) \quad (7.156)$$

$$= (0.45)^{n-k} (0.05)^k \quad (7.157)$$

$$= \left(\frac{1}{2}\right)^n (1-p)^{n-k} p^k. \quad (7.158)$$

An alternative way at looking at this probability is to look at the binary symmetric channel as in additive channel $Y = X \oplus Z$, where Z is a binary random variable that is equal to 1 with probability p , and is independent of X . In this case,

$$p(x^n, y^n) = p(x^n) p(y^n | x^n) \quad (7.159)$$

$$= p(x^n) p(z^n | x^n) \quad (7.160)$$

$$= p(x^n) p(z^n) \quad (7.161)$$

$$= \left(\frac{1}{2}\right)^n (1-p)^{n-k} p^k. \quad (7.162)$$

Show that the condition that (x^n, y^n) being jointly typical is equivalent to the condition that x^n is typical and $z^n = y^n - x^n$ is typical.

- (d) We now calculate the size of $A_\epsilon^{(n)}(Z)$ for $n = 25$ and $\epsilon = 0.2$. As in Problem 3.13, here is a table of the probabilities and numbers of sequences with k ones:

k	$\binom{n}{k}$	$\binom{n}{k} p^k (1-p)^{n-k}$	$-\frac{1}{n} \log p(x^n)$
0	1	0.071790	0.152003
1	25	0.199416	0.278800
2	300	0.265888	0.405597
3	2300	0.226497	0.532394
4	12650	0.138415	0.659191
5	53130	0.064594	0.785988
6	177100	0.023924	0.912785
7	480700	0.007215	1.039582
8	1081575	0.001804	1.166379
9	2042975	0.000379	1.293176
10	3268760	0.000067	1.419973
11	4457400	0.000010	1.546770
12	5200300	0.000001	1.673567

[Sequences with more than 12 ones are omitted since their total probability is negligible (and they are not in the typical set).] What is the size of the set $A_\epsilon^{(n)}(Z)$?

- (e) Now consider random coding for the channel, as in the proof of the channel coding theorem. Assume that 2^{nR} codewords $X^n(1), X^n(2), \dots, X^n(2^{nR})$ are chosen uniformly over the 2^n possible binary sequences of length n . One of these codewords is chosen and sent over the channel. The receiver looks at the received sequence and tries to find a codeword in the code that is jointly typical with the received sequence. As argued above, this corresponds to finding a codeword $X^n(i)$ such that $Y^n - X^n(i) \in A_\epsilon^{(n)}(Z)$. For a fixed codeword $x^n(i)$, what is the probability that the received sequence Y^n is such that $(x^n(i), Y^n)$ is jointly typical?
- (f) Now consider a particular received sequence $y^n = 000000\dots 0$, say. Assume that we choose a sequence X^n at random, uniformly distributed among all the 2^n possible binary

n -sequences. What is the probability that the chosen sequence is jointly typical with this y^n ? [Hint: This is the probability of all sequences x^n such that $y^n - x^n \in A_\epsilon^{(n)}(Z)$.]

- (g) Now consider a code with $2^9 = 512$ codewords of length 12 chosen at random, uniformly distributed among all the 2^n sequences of length $n = 25$. One of these codewords, say the one corresponding to $i = 1$, is chosen and sent over the channel. As calculated in part (e), the received sequence, with high probability, is jointly typical with the codeword that was sent. What is the probability that one or more of the other codewords (which were chosen at random, independent of the sent codeword) is jointly typical with the received sequence? [Hint: You could use the union bound, but you could also calculate this probability exactly, using the result of part (f) and the independence of the codewords.]
- (h) Given that a particular codeword was sent, the probability of error (averaged over the probability distribution of the channel and over the random choice of other codewords) can be written as

$$P(\text{Error} \mid x^n(1) \text{ sent}) = \sum_{y^n: y^n \text{ causes error}} p(y^n \mid x^n(1)). \quad (7.163)$$

There are two kinds of error: the first occurs if the received sequence y^n is not jointly typical with the transmitted codeword, and the second occurs if there is another codeword jointly typical with the received sequence. Using the result of the preceding parts, calculate this probability of error. By the symmetry of the random coding argument, this does not depend on which codeword was sent.

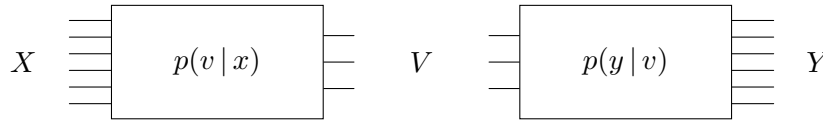
The calculations above show that average probability of error for a random code with 512 codewords of length 25 over the binary symmetric channel of crossover probability 0.1 is about 0.34. This seems quite high, but the reason for this is that the value of ϵ that we have chosen is too large. By choosing a smaller ϵ and a larger n in the definitions of $A_\epsilon^{(n)}$, we can get the probability of error to be as small as we want as long as the rate of the code is less than $I(X; Y) - 3\epsilon$.

Also note that the decoding procedure described in the problem is not optimal. The optimal decoding procedure is maximum likelihood (i.e., to choose the codeword that is closest to the received sequence). It is possible to calculate the average probability of error for a random code for which the decoding is based on an approximation to maximum likelihood decoding, where we decode a received sequence to the unique codeword that differs from the received sequence in ≤ 4 bits, and declare an error otherwise. The only difference with the jointly typical decoding described above is that in the case when the codeword is equal to the received sequence! The average probability of error for this decoding scheme can be shown to be about 0.285.

Problem HW7.2 (C&T 7.16) *Encoder and decoder as part of the channel.* Consider a binary symmetric channel with crossover probability 0.1. A possible coding scheme for this channel with two codewords of length 3 is to encode message a_1 as 000 and a_2 as 111. With this coding scheme, we can consider the combination of encoder, channel, and decoder as forming a new BSC, with two inputs a_1 and a_2 and two outputs a_1 and a_2 .

- Calculate the crossover probability of this channel.
- What is the capacity of this channel in bits per transmission of the original channel?
- What is the capacity of the original BSC with crossover probability 0.1?
- Prove a general result that for any channel, considering the encoder, channel, and decoder together as a new channel from messages to estimated messages will not increase the capacity in bits per transmission of the original channel.

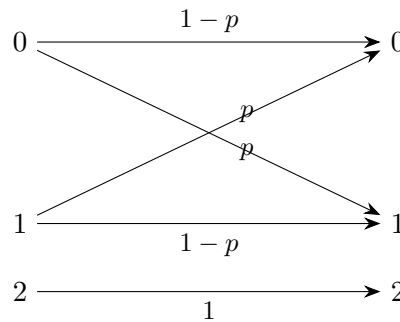
Problem HW7.3 (C&T 7.25) Bottleneck channel. Suppose that a signal $X \in \mathcal{X} = \{1, 2, \dots, m\}$ goes through an intervening transition $X \rightarrow V \rightarrow Y$:



where $x = \{1, 2, \dots, m\}$, $y = \{1, 2, \dots, m\}$, and $v = \{1, 2, \dots, k\}$. Here $p(v|x)$ and $p(y|v)$ are arbitrary and the channel has transition probability $p(y|x) = \sum_v p(v|x)p(y|v)$. Show that $C \leq \log k$.

Problem HW7.4 (C&T 7.28) Choice of channels. Find the capacity C of the union of two channels $(\mathcal{X}_1, p_1(y_1|x_1), \mathcal{Y}_1)$ and $(\mathcal{X}_2, p_2(y_2|x_2), \mathcal{Y}_2)$, where at each time, one can send a symbol over channel 1 or channel 2 but not both. Assume that the output alphabets are distinct and do not intersect.

- Show that $2^C = 2^{C_1} + 2^{C_2}$. Thus, 2^C is the effective alphabet size of a channel with capacity C .
- Compare with Problem 2.10 where $2^H = 2^{H_1} + 2^{H_2}$, and interpret part (a) in terms of the effective number of noise-free symbols.
- Use the above result to calculate the capacity of the following channel.



Problem HW7.5 (C&T 7.36) Channel with memory. Consider the discrete memoryless channel $Y_i = Z_i X_i$ with input alphabet $X_i \in \{-1, 1\}$.

- What is the capacity of this channel when $\{Z_i\}$ is i.i.d. with

$$Z_i = \begin{cases} 1, & p = 0.5 \\ -1, & p = 0.5? \end{cases} \quad (7.168)$$

Now consider the channel with memory. Before transmission begins, Z is randomly chosen and fixed for all time. Thus, $Y_i = Z X_i$.

- What is the capacity if

$$Z = \begin{cases} 1, & p = 0.5 \\ -1, & p = 0.5? \end{cases} \quad (7.169)$$

Problem HW7.6 Random codebooks on the mod-2 additive channel. Consider a channel defined as $Y = (X + E) \bmod 2$, where E is Bernoulli($p = 0.1$) with alphabet $\{0, 1\}$; X is the input symbol and Y is the output symbol. Consider a set of randomly generated codebooks of length 10.

- (a) Suppose we choose a rate $R = 0.4$. Is this choice making sense? If no, choose your own. Construct a single codebook of length $n = 10$.
- (b) Design a maximum-likelihood decision receiver for your codebook.
- (c) Now consider we use Eq. (7.66) in the textbook. How many different codebooks can we obtain?
- (d) Calculate $P(C)$ exactly in your case.
- (e) Show in your case how $P(E) = P(E | W = 1)$.
- (f) Would your ML receiver produce $P(E)$ smaller than p in an average sense? Prove. Also show that $P(E)$ is getting smaller as n increases.